Fox–Wright Orthogonal Polynomials and the Fractional Riccati–Müntz–Padé Theorem: A New Family in Non-Hermitian Approximation Theory

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May 2026

Abstract

We identify and characterize the orthogonal polynomial sequence canonically associated with the Fractional Riccati–Müntz–Padé theorem [Crowley2026]. The polynomials arise as the denominators of the Padé approximants built from the Müntz–Tau moment sequence $\{a(k,u)\}_{k\geq 0}$ of the fractional Riccati equation $D^{\mu} y = P(u) + Q(u)\, y + R(u)\, y^2$, and are orthogonal with respect to a quasi-definite moment functional whose weights are modulated by the Fox--Wright function $\Psi_{\mu}(z) = \sum_{k\geq 0} \Gamma\,(k\,\mu+1)\,/\,\Gamma\,(k\,\mu+\mu+1)\cdot z^k$.

We place these polynomials precisely within the hierarchy of known orthogonal families: they belong to the Nuttall–Stahl class of non-Hermitian orthogonal polynomials [NuttallSingh1977, Stahl1985I, Stahl1985II] on the Stahl minimal-capacity compact $K_A(u)$, form a regular sequence in the sense of Stahl–Totik [StahlTotik1992], and in the $\mu = 1$ special case reduce to Chebotarëv–Jacobi polynomials studied by Barhoumi–Yattselev [BarhoumiYattselev2023]. For $\mu \in (0,1)$ they constitute a genuinely new subfamily – *Fox–Wright non-Hermitian orthogonal polynomials* – whose weight is a Fox–Wright-deformed Jacobi-type function on a transcendental arc system, and whose asymptotic recurrence coefficients are determined by the equilibrium measure of $K_A(u)$ via the Gonchar–Rakhmanov S-property [GoncharRakhmanov1987]. The Müntz–Szász density theorem [MuntzSzasz1914, Moragues2018] provides the underlying qualitative completeness backdrop, while the Chebyshev–Wheeler algorithm [Wheeler1974, Gragg1972] provides the constructive computational realization.

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1 Introduction

The Fractional Riccati–Müntz–Padé theorem [Crowley2026] establishes that the unique solution of the fractional Riccati equation

$$D^{\mu} y(t,u) = P(u) + Q(u)\ y(t,u) + R(u)\ y(t,u)^2 \tag{1}$$

$$I^{1-\mu} y(0,u) = 0 \tag{2}$$

with polynomial coefficients $P, Q, R \in \mathbb{C}[u]$, order $\mu \in (0,1]$, and Riemann–Liouville calculus, is recovered in logarithmic capacity on the complement of Stahl's compact $K_A(u)$ by the near-diagonal Müntz–Padé approximants

$$y_M(t,u) = t^{\mu} \frac{N_M(t^{\mu},u)}{D_M(t^{\mu},u)} \tag{3}$$

Central to the proof is the identification of the Padé denominator sequence $\{D_M\}$ with the orthogonal polynomial sequence of the moment functional $\mathcal{L}[x^k] = a(k,u)$, where $\{a(k, u)\}$ are the Müntz–Tau coefficients satisfying an explicit Fox–Wright-weighted recurrence [Crowley2026]. The present article characterizes this orthogonal polynomial sequence in depth: its algebraic structure, its place in the classification of orthogonal polynomial families, its asymptotic behavior, and its relationship to classical and modern results in non-Hermitian approximation theory.

The main contribution is the identification of these polynomials as *Fox–Wright non-Hermitian orthogonal polynomials* (FW-NHOPs) – a new subfamily of the Nuttall–Stahl class parametrized by $\mu \in (0,1)$ that interpolates between the classical algebraic (Chebotarëv–Jacobi) case at $\mu = 1$ and a transcendental Fox–Wright-weighted limit as $\mu \to 0^+$. The Müntz–Szász condition $\sum 1/(k\,\mu) = \infty$ for all $\mu \in (0,1]$ guarantees the underlying density [Moragues2018], while the Chebyshev–Wheeler algorithm [Wheeler1974] provides numerically stable computation of the recurrence coefficients.

2 The Moment Functional and its Fox–Wright Structure

2.1 The Müntz–Tau Moments

The fractional Riccati equation (1), recast as the Volterra integral equation $y = I^\mu [P + Q\,y + R\,y^2]$ via the Riemann–Liouville composition formula [SKM1993], admits a unique formal power series solution in the Müntz basis $\{t^{(k+1)\mu}\}_{k\geq 0}$ [Crowley2026]:

$$y(t,u) = \sum_{k\geq 0} a(k,u)\ t^{(k+1)\mu} \tag{4}$$

with $a(k,\ \cdot\ ) \in \mathbb{C}[u]$ satisfying the recurrence

$$a(0,u) = P(u)\ \psi_0 \tag{5}$$

$$a(k,u) = \psi_k\left(Q(u)\ a(k-1,u) + R(u) \sum_{j+\ell=k-1} a(j,u)\ a(\ell,u)\right) \forall k \geq 1 \tag{6}$$

where the Fox–Wright weights are

$$\psi_k := \frac{\Gamma\,(k\,\mu+1)}{\Gamma\,(k\,\mu+\mu+1)} \quad \forall k \geq 0 \tag{7}$$

The sequence $\{\psi_k\}$ is precisely the sequence of eigenvalues of the fractional integral I^μ on the Müntz monomial basis: $I^\mu\,t^{(k+1)\mu} = \psi_{k+1}\ t^{(k+2)\mu}$ [Crowley2026]. By Stirling's formula, $\psi_k \sim (k\,\mu)^{-\mu}$ as $k \to \infty$, and $\Psi_\mu(z) = \sum_{k\geq 0} \psi_k\,z^k$ has radius of convergence exactly 1.

2.2 Gevrey-1 Growth and Formal vs. Analytic Structure

The generating function $A(z,u) = \sum_{k\geq 0} a(k,u)\ z^k$ lives in $\mathbb{C}[u][[z]]$ as a formal power series. By an explicit induction on the recurrence (6), the moments satisfy the Gevrey-1 bound [Crowley2026]:

$$|a(k,u)| \leq K_G(u)\ C_G(u)^k\ k! \tag{8}$$

with computable constants K_G, C_G depending on $|P(u)|$, $|Q(u)|$, $|R(u)|$, and $\psi_0 = 1/\Gamma(\mu + 1)$. The factorial growth means the radius of convergence of A is zero for generic parameters – A diverges everywhere except at the origin.

This divergence is not a defect: it is the signature of the meromorphic nature of A. Via a Mellin–Barnes integral representation [Crowley2026]

$$A(z,u) = \frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty} M_{\Psi_\mu}(s)\ \hat{B}_R(-s,u)\ (-z)^{-s}\ d\,s \tag{9}$$

where

$$M_{\Psi_\mu}(s) = \Gamma(s)\,\Gamma(1-s)\,\Gamma(1-s\,\mu)/\Gamma(\mu+1-s\,\mu) \tag{10}$$

is the Mellin transform of $\Psi_\mu(-x)$, the formal series A is promoted to a meromorphic function on the universal cover $\tilde{X}$ of $\mathbb{C}\setminus\{0\}$ minus a minimal-capacity compact $K_A(u)$ – Stahl's compact.

2.3 The Hadamard Fixed-Point Equation

Define

$$B[A](z,u) := P(u) + z\,Q(u)\,A(z,u) + z\,R(u)\,A(z,u)^2 \in \mathbb{C}[u][[z]] \tag{11}$$

Then the unique formal power series satisfying the Müntz–Tau recurrence is characterized by the Hadamard fixed-point equation [Crowley2026]:

$$A(z,u) = \Psi_\mu(z) \odot B[A](z,u), \tag{12}$$

where $\odot$ denotes the Hadamard (coefficient-wise) product: $[z^k]\,(F\odot G) = f_k\,g_k$. This identity holds in $\mathbb{C}[u][[z]]$ and is proved by direct coefficient extraction [Crowley2026]: at degree k,

$$[z^k]\,B[A] = Q\ a\,(k-1) + R\sum_{j+\ell=k-1} a(j)\,a(\ell) \tag{13}$$

and multiplying by ψ_k recovers $a(k)$ exactly from (6).

3 The Moment Functional and Its Quasi-Definiteness

Definition 1. *[Fox–Wright moment functional] The* Fox–Wright moment functional *associated to* (1) *is the* $\mathbb{C}$*-linear map* $\mathcal{L}: \mathbb{C}[x] \to \mathbb{C}[u]$ *defined by*

$$\mathcal{L}[x^k] := a(k, u) \forall \forall k \geq 0 \tag{14}$$

extended by linearity. The Hankel determinants are

$$H_n(u) := \det(a(i+j, u))_{0 \leq i, j \leq n-1} \in \mathbb{C}[u] \tag{15}$$

The functional is quasi-definite *if* $H_n(u) \neq 0$ *for all* $n \geq 1$ *(as elements of* $\mathbb{C}[u]$*, i.e., not identically zero).*

Remark 2. [Generic quasi-definiteness] By the Müntz–Tau recurrence (5)–(6), each $a(k, u)$ is a polynomial in u of degree increasing with k. The Hankel determinant $H_n(u)$ is a polynomial in u that vanishes only on a proper algebraic subvariety of parameter space; generically, $H_n(u) \neq 0$. The recurrence coefficients $\alpha(k, u)$ and $\beta(k, u)$ produced by the Chebyshev–Wheeler algorithm [Wheeler1974, Gragg1972] are rational functions of u well-defined off this subvariety.

Under quasi-definiteness, the Chebyshev–Wheeler algorithm [Wheeler1974, Gautschi1982] produces the unique monic polynomial sequence $\{P_M(x, u)\}_{M \geq 0}$ satisfying

$$\mathcal{L}[x^k P_M(x, u)] = 0 \qquad \forall k = 0, 1, \ldots, M-1 \tag{16}$$

via the three-term recurrence

$$P_{M+1}(x, u) = (x - \alpha(M, u))\, P_M(x, u) - \beta(M, u)\, P_{M-1}(x, u) \tag{17}$$

The Padé denominator of $A(z, u)$ at order M is the polynomial

$$D_M(z, u) = z^M P_M(z^{-1}, u) \tag{18}$$

the reciprocal transform of P_M [Crowley2026, Gragg1972].

4 Classification: The Fox–Wright Non-Hermitian Orthogonal Polynomials

4.1 The Hierarchy of Orthogonal Polynomial Families

We locate $\{P_M(\cdot, u)\}$ within the classification of orthogonal polynomial sequences:

1. **Classical orthogonal polynomials** (Jacobi, Laguerre, Hermite) are orthogonal with respect to a positive measure on $\mathbb{R}$ with explicit weight function. The moments grow at most polynomially. The recurrence coefficients $\alpha(k)$ and $\beta(k)$ are rational functions of k with known explicit forms. *Our polynomials do not belong to this class*: by (8), the moments grow like $k!$, ruling out any positive measure on $\mathbb{R}$ [StahlTotik1992].

2. **Favard class / positive-measure OPs** are orthogonal with respect to a positive Borel measure on $\mathbb{C}$ with compact support. *Our polynomials do not belong to this class* in general: the moment functional $\mathcal{L}$ is quasi-definite but not positive-definite (the Hankel determinants may have varying signs or be complex-valued for generic $u \in \mathbb{C}$).

3. **Regular OPs in the Stahl–Totik sense** [StahlTotik1992] satisfy the n-th root asymptotics

$$|P_M(z, u)|^{1/M} \to e^{g(z, K_A(u))} \operatorname{cap}(K_A(u)) \quad \text{in capacity on } \mathbb{C} \setminus K_A(u) \tag{19}$$

   where $g(\cdot, K_A(u))$ is the Green function of $\mathbb{C} \setminus K_A(u)$ with pole at infinity, and $\operatorname{cap}(K_A(u))$ is the logarithmic capacity. *Our polynomials belong to this class* by virtue of Stahl's convergence theorem [Stahl1997] as invoked in [Crowley2026].

4. **Nuttall–Stahl non-Hermitian OPs** [NuttallSingh1977, Stahl1985I, Stahl1985II, GoncharRakhmanov1987] are orthogonal on the Stahl minimal-capacity contour K_Φ with a complex-valued weight ρ (the jump of the approximated function across K_Φ):

$$\int_{K_A(u)} z^k P_M(z, u)\ \rho(z, u)\ dz = 0\ \forall k = 0, \dots, M-1 \tag{20}$$

   Our polynomials belong to this class, with $K_A(u)$ the Stahl compact from [Crowley2026] and $\rho(z, u)$ the jump of $A(z, u)$ across $K_A(u)$ – a Fox–Wright-deformed Jacobi-type function described in Section 4.2 below.

4.2 The Weight Function

The jump of $A(z,u)$ across $K_A(u)$ is determined by the algebraic structure of the auxiliary function $F(z,u)$ satisfying the quadratic [Crowley2026]:

$$z\,R(u)\;F^2-(1-z\,Q(u))\;F+P(u)\;\Psi_\mu(z)=0 \tag{21}$$

The discriminant is

$$\Delta_F(z,u)=(1-z\,Q(u))^2-4\,z\,P(u)\,R(u)\;\Psi_\mu(z) \tag{22}$$

and $F(z,u)$ has the closed form

$$F(z,u)=\frac{(1-z\,Q)-\sqrt{\Delta_F}}{2\,z\,R} \tag{23}$$

The jump of F across the branch cut $K_F(u)$ is

$$\rho_F(z,u)=F^+(z,u)-F^-(z,u)=\frac{\sqrt{\Delta_F^+(z,u)}-\sqrt{\Delta_F^-(z,u)}}{2\,z\,R(u)} \tag{24}$$

which is a Fox–Wright-modified Jacobi-type weight: the classical Jacobi weight $\sqrt{(z-e_1)\,(z-e_2)}$ (where e_1,e_2 are the branch points of the algebraic part $(1-z\,Q)^2-4\,z\,P\,R$) is deformed by the factor $\Psi_\mu(z)$ in (22). Since A and F have the same branch locus by the branch-transfer lemma [Crowley2026], $\rho_A=\rho_F$ up to a meromorphic factor, and the non-Hermitian orthogonality (20) holds with weight $\rho(z,u)=\rho_F(z,u)$.

4.3 The Fox–Wright Deformation

The key distinction from all previously named orthogonal polynomial families is the presence of $\Psi_\mu(z)$ in the discriminant (22). For $\mu=1$, one computes

$$\Psi_1(z)=\sum_{k\geq 0}\frac{1}{k+1}\;z^k=-\frac{\log(1-z)}{z} \tag{25}$$

which is analytic in $|z|<1$ but has a logarithmic branch point at $z=1$. Thus even in the simplest non-classical case, Δ_F acquires a transcendental factor. For $\mu\in(0,1)$, Ψ_μ is a Fox–Wright ${}_1\Psi_1$ function (see [Wright1935, Gorenflo2014]):

$$\Psi_\mu(z)={}_1\Psi_1\!\left(\begin{array}{c}(1,\mu)\\(1+\mu,\mu)\end{array};z\right)=\sum_{k\geq 0}\frac{\Gamma(k\,\mu+1)}{\Gamma(k\,\mu+\mu+1)}\;z^k \tag{26}$$

with Mellin transform

$$M_{\Psi_\mu}(s) = \frac{\Gamma(s)\,\Gamma\,(1-s)\,\Gamma\,(1-s\,\mu)}{\Gamma\,(\mu+1-s\,\mu)} \tag{27}$$

meromorphic on $\mathbb{C}$ [Crowley2026]. The branch points of $\sqrt{\Delta_F}$ are therefore the zeros of (22), which include both the zeros of the polynomial factor $(1-z\,Q)^2-4\,z\,PR\cdot\psi_0$ (near $z=0$) and the singularities of Ψ_μ on the circle $|z|=1$, projected under the Stahl–Chebotarëv minimization [Stahl1985I, Stahl1985II].

5 Asymptotic Theory

5.1 The S-Property and Stahl's Compact

The Stahl compact $K_A(u)$ satisfies the *S-property* of Gonchar–Rakhmanov [GoncharRakhmanov1987]: the normal derivative of the Green function $g(z, K_A(u))$ from the two sides of $K_A(u)$ are equal

$$\frac{\partial g}{\partial n^+}(z, K_A(u)) = \frac{\partial g}{\partial n^-}(z, K_A(u)) \;\; \forall z \in K_A(u) \setminus \partial K_A(u) \tag{28}$$

This characterizes $K_A(u)$ as the support of the equilibrium measure that balances the logarithmic energy functional [Ransford1995]:

$$\nu_{K_A(u)} = \arg\min_{\nu} I[\nu] \tag{29}$$

$$I[\nu] := \iint \log\frac{1}{|z-w|}\; d\,\nu(z)\; d\,\nu(w) \tag{30}$$

5.2 Zero Condensation: Nuttall–Stahl

The empirical zero-counting measure of the orthogonal polynomials

$$\nu_M := \frac{1}{M} \sum_{P_M(\lambda,u)=0} \delta_\lambda \tag{31}$$

converges weak-$*$ to $\nu_{K_A(u)}$ as $M\to\infty$, the equilibrium measure of the Stahl compact [Crowley2026, Stahl1997]. This is the Nuttall–Stahl zero-condensation theorem, which in the present context reads: the zeros of $\{P_M(\cdot,u)\}_{M\geq 1}$ are dense in $K_A(u)$ and equidistribute according to $\nu_{K_A(u)}$.

5.3 Recurrence Coefficient Asymptotics

Under the regularity hypothesis (19), the recurrence coefficients of (17) satisfy [StahlTotik1992]:

$$\alpha(k,u) \longrightarrow \alpha^*(u) \tag{32}$$

$$\beta(k,u) \longrightarrow \beta^*(u) \tag{33}$$

as $k \to \infty$, where $\alpha^*(u)$ and $\beta^*(u)$ are determined by the conformal map $\Phi: \mathbb{C} \setminus K_A(u) \to \mathbb{C} \setminus \bar{\mathbb{D}}$ by the Szegő-type asymptotic formula. Explicitly, the equilibrium measure $\nu_{K_A(u)}$ has density

$$d\nu_{K_A(u)} = (1/2\pi)|\Phi'(z)|\ |dz| \tag{34}$$

on $K_A(u)$, and the asymptotic recurrence coefficients are the first two Laurent coefficients of the conformal map:

$$\Phi(z) = z/\mathrm{cap}(K_A(u)) + \alpha^*(u) + O(1/z). \tag{35}$$

For $K_A(u)$ an interval $[a,b] \subset \mathbb{R}$ (possible for special real parameter values), this reduces to

$$\alpha^*(u) = \tfrac{a+b}{2} \tag{36}$$

$$\beta^*(u) = \tfrac{(b-a)^2}{16} \tag{37}$$

and the $P_M(\cdot,u)$ are asymptotically Chebyshev polynomials for $[a,b]$. For $K_A(u)$ a complex arc, the conformal map is transcendental and no closed-form expression is known in general.

5.4 Convergence: Stahl's Theorem Applied

The Padé approximants N_M/D_M converge to $A(z,u)$ in logarithmic capacity on compact subsets of $\tilde{X} \setminus K_A(u)$:

$$\left|\frac{N_M}{D_M} - A\right|^{1/(2M)} \longrightarrow e^{-g(z,K_A(u))} \quad \text{in capacity} \tag{38}$$

[Stahl1997, Crowley2026]. This implies exponential convergence off $K_A(u)$ with rate governed by the Green function. Since $K_A(u)$ is disjoint from $\mathbb{R}_+$ (the poles of A on the physical time axis are absent by a Volterra-iteration argument [Crowley2026]), the convergence upgrades to uniform convergence:

$$\sup_{t\in[0,T]} |y(t,u) - y_M(t,u)| \longrightarrow 0 \qquad \text{as } M \to \infty. \tag{39}$$

6 The $\mu = 1$ Special Case: Chebotarëv–Jacobi Polynomials

When $\mu = 1$, the Fox–Wright weights reduce to

$$\psi_k = \frac{1}{k+1} \tag{40}$$

and

$$\Psi_1(z) = \frac{-\log(1-z)}{z} \tag{41}$$

The discriminant (22) becomes

$$\Delta_F(z,u)|_{\mu=1} = (1 - z\,Q)^2 + 4\,z\,P R\,\frac{\log(1-z)}{z} \tag{42}$$

The zeros of Δ_F are the solutions to

$$(1 - z\,Q)^2 = -4\,P R \log(1-z) \tag{43}$$

a transcendental equation, but the branch structure of $\sqrt{\Delta_F}$ is still two-sheeted, and the Stahl compact $K_A(u)$ is a finite arc in $\mathbb{C}$ connecting the two branch points of $\sqrt{\Delta_F}$. The orthogonal polynomials in this case are exactly the Padé denominator polynomials for a two-sheeted function with one finite branch cut, the class studied in [BarhoumiYattselev2023] and earlier by Aptekarev–Stahl [Stahl1997].

For the further special case $Q = 0$, $R = -1$, $P = 1$ (which gives the Bernoulli equation at $\mu = 1$), the quadratic (21) simplifies and $K_A(u)$ is a segment $[0, \sigma]$ for σ the first zero of the classical Riccati solution – the $P_M(\cdot, u)$ are then Legendre polynomials shifted to $[0, \sigma]$.

7 The $R = 0$ Oracle: Mittag-Leffler Moments and Laguerre Structure

When $R(u) = 0$, the recurrence (6) linearizes to

$$a(k,u) = \psi_k\,Q(u)^k\,P(u)\,\psi_0^{-1} \cdot \psi_0 \tag{44}$$

giving [Crowley2026]:

$$a(k,u) = P(u)\;\psi_k\;Q(u)^k \tag{45}$$

The moment functional becomes

$$\mathcal{L}[x^k] = P(u)\,\psi_k\,Q(u)^k \tag{46}$$

a Stieltjes-type functional with moments $\{P\psi_k Q^k\}$.

Proposition 3. *In the $R = 0$ case, the moment functional $\mathcal{L}$ has the integral representation*

$$\mathcal{L}[f] = \int_0^\infty f\,(Q\,x)\;w_\mu(x)\;d\,x \tag{47}$$

where

$$w_\mu(x) = P\;\mathcal{L}^{-1}[\psi_k](x) \tag{48}$$

is the inverse Laplace transform of the Fox–Wright function $z \mapsto P\;\Psi_\mu\,(Q\,z)$. Explicitly, w_μ is a one-sided stable density with stability index μ, related to the Mittag-Leffler measure [Grothaus2015].

The $R = 0$ orthogonal polynomials are therefore classical Laguerre-type polynomials for the one-sided stable distribution at index μ. For $\mu = 1$, $w_1(x) = Pe^{-x}$ and the polynomials are standard Laguerre polynomials $L_M^{(0)}$ scaled to $[0, 1/Q]$. For $\mu \neq 1$, they are the orthogonal polynomials of the Mittag-Leffler measure of [Grothaus2015] – a family studied in connection with fractional Brownian motion and anomalous diffusion.

This provides a closed-form unit test for any implementation: the $[M/M]$ Padé approximant to $A(z,u)|_{R=0} = P/(1 - Q\,z)$ must reproduce the M-th Padé approximant to $E_{\mu,1}\,(Q\,t^\mu)$ exactly, and the Wheeler algorithm must recover Laguerre recurrence coefficients asymptotically [Crowley2026].

8 The Müntz–Szász Backbone

The Müntz–Szász theorem [MuntzSzasz1914, Moragues2018] provides the qualitative underpinning of the entire convergence theory. The Müntz exponents used in the expansion (4) are $\lambda_k = (k+1)\,\mu$, and

$$\sum_{k=0}^{\infty} \frac{1}{\lambda_k} = \frac{1}{\mu} \sum_{k=0}^{\infty} \frac{1}{k+1} = \infty \tag{49}$$

for every $\mu \in (0, 1]$. By Müntz–Szász, the system $\{t^{(k+1)\mu}\}$ is therefore dense in $\{f \in C([0, T]) \colon f(0) = 0\}$ for all $T > 0$.

Three implications follow immediately:

1. **No coefficient can be discarded.** Any subsequence $\{a(k_j, u)\}$ with $\sum 1/k_j < \infty$ generates a Padé table that cannot uniformly recover $y(t,u)$ on $[0, T]$. All moments are indispensable.

2. **The Müntz rational approximants have maximum approximation power.** Since the exponents satisfy (49), the Müntz rationals $\{t^\mu\,N_M(t^\mu, u)/D_M\,(t^\mu, u)\}$ form a complete approximating family in $C_0([0, T])$.

3. **Uniform convergence on $\mathbb{R}_+$ is the canonical upgrade.** The Stahl convergence-in-capacity result on the complement of $K_A(u)$, combined with the absence of poles on $\mathbb{R}_+$ (Lemma 14 of [Crowley2026]), is the quantitative realization of the qualitative Müntz–Szász density – the divergence of $\sum 1/\lambda_k$ is the engine, and the Fox–Wright-weighted Padé table is the mechanism.

9 Computational Realization: The Chebyshev–Wheeler Algorithm

The three-term recurrence coefficients $\alpha(k,u)$ and $\beta(k,u)$ are computed from the moments $\{a(k,u)\}$ by the Chebyshev–Wheeler algorithm [Wheeler1974, Gautschi1982], which operates on an auxiliary array $\sigma(k,\ell;u)$ initialized by

$$\sigma(0,\ell;u) = a(\ell,u) \tag{50}$$

and evolved by

$$\sigma(k,\ell;u) = \sigma(k-1,\ell+1;u) - \alpha(k-1,u)\ \sigma(k-1,\ell;u) - \beta(k-1,u)\ \sigma(k-2,\ell;u) \tag{51}$$

The recurrence coefficients are

$$\alpha(k,u) = \frac{\sigma(k,k+1;u)}{\sigma(k,k;u)} - \frac{\sigma(k-1,k;u)}{\sigma(k-1,k-1;u)} \tag{52}$$

$$\beta(k,u) = \frac{\sigma(k,k;u)}{\sigma(k-1,k-1;u)} \tag{53}$$

Under quasi-definiteness, $\sigma(k,k;u) \neq 0$ for all k and the algorithm is well-defined. The orthonormality of the resulting polynomials is guaranteed by Wheeler's theorem [Wheeler1974]:

$$\mathcal{L}\left[\pi_j\ \pi_k\right] = \delta_{jk} \prod_{i=0}^{k} \beta(i,u) \tag{54}$$

where π_k are the monic orthogonal polynomials produced by (17).

Numerical stability. The Chebyshev–Wheeler algorithm for the Fox–Wright moment functional is subject to the same conditioning issues as the general modified-moment algorithm [Gautschi1994, Gautschi1998]: the map from moments to recurrence coefficients is a nonlinear map that can be ill-conditioned when the moments grow rapidly. The Gevrey-1 bound (8) implies moments of size $\sim K_G\, C_G^k\, k!$, and working in extended precision (as in the ARB library [Johansson2017]) is recommended for large M.

10 Summary: Classification of the FW-NHOP Family

Property	Value/Status	Reference
Moment growth	Gevrey-1: $\lvert a(k)\rvert \leq K\,C^k\,k!$	[Crowley2026]
Moment functional	Quasi-definite, not positive	[Crowley2026]
Orthogonality	Non-Hermitian on $K_A(u)$	[NuttallSingh1977]
Weight	Fox–Wright-deformed Jacobi	[Wright1935, Crowley2026]
Zero distribution	Condenses to $\nu_{K_A(u)}$	[Stahl1997, Crowley2026]
Regularity	Stahl–Totik regular	[StahlTotik1992]
S-property	Gonchar–Rakhmanov	[GoncharRakhmanov1987]
$\mu = 1$ reduction	Chebotarëv–Jacobi	[BarhoumiYattselev2023]
$R = 0$ reduction	Mittag-Leffler / Laguerre-type	[Grothaus2015]
Computation	Chebyshev–Wheeler	[Wheeler1974, Gautschi1982]
Convergence	Stahl cap. + uniform on $\mathbb{R}_+$	[Stahl1997, Crowley2026]
Completeness	Müntz–Szász	[Moragues2018]

Table 1. Properties of the Fox–Wright non-Hermitian orthogonal polynomials (FW-NHOPs) associated to the fractional Riccati equation (1).

The Fox–Wright non-Hermitian orthogonal polynomials form a one-parameter family indexed by $\mu \in (0,1]$ that interpolates between:

- $\mu = 1$: classical (logarithmically-deformed) Chebotarëv–Jacobi polynomials on a two-arc system [BarhoumiYattselev2023];
- $\mu \to 0^+$: a limiting regime where $\psi_k \to 1$ for all k (by $\Gamma(1)/\Gamma(1) = 1$) and the Fox–Wright weight collapses to a standard algebraic weight – the polynomials approach classical Padé denominators for the bare quadratic $F_0(z,u)$ with $\Psi_0 \equiv 1$.

Bibliography

[Crowley2026] S. Crowley, *The Fractional Riccati–Müntz–Padé Theorem*, preprint (2026).

[SKM1993] S. G. Samko, A. A. Kilbas, O. I. Marichev, *Fractional Integrals and Derivatives: Theory and Applications*, Gordon and Breach, 1993.

[Stahl1985I] H. Stahl, *Extremal domains associated with an analytic function I*, Constr. Approx. **1** (1985), 89–110.

[Stahl1985II] H. Stahl, *Extremal domains associated with an analytic function II*, Constr. Approx. **1** (1985), 111–137.

[Stahl1997] H. Stahl, *The convergence of Padé approximants to functions with branch points*, J. Approx. Theory **91** (1997), 139–204.

[StahlTotik1992] H. Stahl, V. Totik, *General Orthogonal Polynomials*, Encyclopedia of Mathematics and its Applications **43**, Cambridge University Press, 1992.

[GoncharRakhmanov1987] A. A. Gonchar, E. A. Rakhmanov, *Equilibrium distributions and degree of rational approximation of analytic functions*, Mat. Sb. **134**(176) (1987), 306–352; English transl. Math. USSR-Sb. **62** (1989), 305–348.

[NuttallSingh1977] J. Nuttall, S. R. Singh, *Orthogonal polynomials and Padé approximants associated with a system of arcs*, J. Approx. Theory **21** (1977), 1–42.

[BarhoumiYattselev2023] A. B. Barhoumi, M. L. Yattselev, *Non-Hermitian orthogonal polynomials on a trefoil*, arXiv:2303.17037 (2023).

[Wheeler1974] J. C. Wheeler, *Modified moments and Gaussian quadratures*, Rocky Mountain J. Math. **4** (1974), 287–296.

[Gragg1972] W. B. Gragg, *The Padé table and its relation to certain algorithms of numerical analysis*, SIAM Review **14** (1972), 1–62.

[Gautschi1982] W. Gautschi, *On generating orthogonal polynomials*, SIAM J. Sci. Stat. Comput. **3** (1982), 289–317.

[Gautschi1994] W. Gautschi, *Orthogonal polynomials: applications and computation*, Acta Numerica **5** (1996), 45–119.

[Gautschi1998] W. Gautschi, L. Gori, M. L. Lo Cascio, *Quadrature rules for rational functions*, Numer. Math. **86** (2000), 617–633.

[MuntzSzasz1914] Ch. H. Müntz, *Über den Approximationssatz von Weierstrass*, H. A. Schwarz Festschrift, Berlin, 1914, 303–312.

[Moragues2018] A. Ferré Moragues, *What is... the Müntz–Szász Theorem?*, expository note (2018).

[Wright1935] E. M. Wright, *The asymptotic expansion of the generalized Bessel function*, Proc. London Math. Soc. (2) **38** (1935), 257–270.

[Gorenflo2014] R. Gorenflo, A. A. Kilbas, F. Mainardi, S. V. Rogosin, *Mittag-Leffler Functions, Related Topics and Applications*, Springer, 2014.

[Grothaus2015] M. Grothaus, F. Jahnert, *Mittag-Leffler analysis I: Construction and characterization*, J. Funct. Anal. **268** (2015), 1876–1903.

[Ransford1995] T. Ransford, *Potential Theory in the Complex Plane*, London Math. Soc. Student Texts **28**, Cambridge Univ. Press, 1995.

[Johansson2017] F. Johansson, *Arb: efficient arbitrary-precision midpoint-radius interval arithmetic*, IEEE Trans. Comput. **66** (2017), 1281–1292.